Swimming Stroke Video Classification using a Convolutional Neural Network

Myles Felt Elinor Frost Jack Stewart Isaac Viviano

December 17, 2024

Abstract

Competitive swimming is a growing industry, and athletes are continuously pushing the boundaries of human athletic capability. The integration of video analysis technology to the sport may allow athletes to achieve even higher performance. We address the problem of stroke classification in video data by training a convolutional neural network. We utilize video data from `SwimXYZ`–a data set of synthesized videos depicting swimmers performing the four competitive strokes–and the `MC3_18` architecture to develop this model. At peak performance, our model was able to correctly classify breaststroke and freestyle from video frames with a validation accuracy of 79.5%. We provide suggestions for future work to improve the model's performance, and we expand on the numerous applications for which our model can be used.

1 Introduction

The world of competitive swimming is growing in interest and revenue, with the Olympic year bringing more viewership to the sport than ever. Additionally, athletes are performing at higher levels than ever before; between December 10th and 15th of 2024, there were an unprecedented 30 world record performances at the World Aquatics Swimming Championships in Budapest [1]. For swimmers to continue testing the limits of athletic performance in swimming, advances in technology-driven stroke analysis and visual feedback tools are critical.

Competitive swimming involves four strokes, each of which is designed to propel the swimmer through the water in different ways. We address the problem of identifying the stroke being swum at certain frames in a video. This specific problem has several applications in the world of swimming. Identifying certain aspects of a stroke in a video may allow officials to automate the disqualification process (when swimmers utilize movement during a stroke that is not allowed). Additionally, stroke identification from videos of training sessions may streamline the process of providing technical feedback to swimmers.

To address the problem of identifying strokes in video frames, we trained a Convolutional Neural Network (CNN) using data from the `SwimXYZ` data set, which contains thousands of labeled videos of simulated swimmers performing the four different strokes in various environments. The network is structured using the pre-defined specifications of the `MC3_18` video classification model in PyTorch. We use the process described by [2] to build, train, and track performance of the model. Our CNN was trained for 20 epochs with a learning rate of 0.001.

Due to time and resource constraints, we reduced our classification task to two strokes: freestyle and breaststroke. Additionally, we used a small subset of the data and restricted our training set to videos depicting only a side angle. Our CNN had a modest overall performance in classifying these two strokes, with a peak validation accuracy of 79.5% at epoch 17. We conclude by providing several recommendations for future work. These include training a model to include butterfly and backstroke, utilizing a greater volume of data from `SwimXYZ` for training, and adjusting the hyperparameters of the model to maximize prediction accuracy.

2 Background and Related Work

2.1 Stroke Differentiation

Competitive swimming strokes all have unique characteristics and performance-related statistics. To understand our task of classification, it is important to understand how different strokes are performed. There are four swimming strokes, each of which differ in the techniques used for propulsion, and in the swimmer's body and head position at any given time. These four strokes can be further split into the long-axis and short-axis strokes.

Long-axis strokes are freestyle (also known as the front-crawl) and backstroke. In these strokes, the body and head remain in a horizontal line on the surface of the water with most body movement consisting of rotating about this center horizontal line. Both long-axis strokes share a kick in which each leg alternates positions and there is no knee bend. Additionally, they consist of single-arm movement, where the arms alternate to pull water. Backstroke and freestyle differ in the way the swimmer's body is facing. In freestyle, the swimmer's face is in the water facing down, and the swimmer's back faces the sky or ceiling. In backstroke, the swimmer's face is toward the sky or ceiling, and the back is facing the bottom of the pool.

Breaststroke and butterfly are the short-axis strokes. While the long-axis strokes keep a consistent body position along the length of the body, the short-axis strokes have much more pronounced movements along the body. Butterfly features a kick which involves the entire lower body and much of the torso making an undulating up and down motion, while the arms rotate in unison around the water. Breaststroke can be best described as a frog motion, with the kick involving the legs making a "V" shape before snapping back together. The upper body in breaststroke is lifted out of the water while the arms are shot forward and then pull water.

2.2 Related Work

Others have attempted the task of classifying swimming strokes based on real-world data, which can take many forms. Studies that collect data for various tasks related to competitive swimming often employ one of two devices: wearable technology or video/timing mechanisms. Common wearable technology include smartwatches and biometric sensors (e.g. heart-rate monitors) [3]. These devices are worn by the swimmer throughout a swim workout and capture data using various sensors. Smartwatches often have gyroscopes, light sensors, accelerometers, and pressure sensors, which can be used to collect data for health purposes. [4] collected data using these smartwatch features in their study on the stroke classification task. Smartwatches are also used for collecting biometric data (e.g. heart rate, respiration). Wearables on the head, such as the Attitude and Head Referencing System studied by [5], utilize data such as body position, body roll, and breathing patterns to provide technical feedback for improving a swimmer's stroke. Machine learning tasks related to competitive swimming also utilize video and timing mechanisms. [6] capture and isolate video frames of a swimmer and use computer vision to analyze body positioning for optimizing stroke efficiency. [7] use data from cameras and timing mechanisms to find and analyze split times, stroke rates, distance traveled per stroke, and speed at different parts of a swim.

Competitive swimming lends itself to many machine learning tasks, including our selected task of stroke classification. This problem has previously been studied by [4]; however, the researchers approach this task using data from wearable-technology, rather than video and image data. The researchers used sensor data from a smartwatch to classify strokes and distance swam during a pool workout. Other related machine-learning applications have focused on evaluating swimming techniques and race strategy to help athletes swim faster and more efficiently [7, 5]. Machine learning research has also used race results from online databases to study trends for various prediction and classification tasks, including predicting times based on factors such as gender and age [8].

3 Problem

In this project, we address the problem of identifying competitive swimming strokes based on video frames of swimmers performing those strokes. During a training session, a competitive swimmer typically performs a mix of the different strokes at different times. Programs with the capability of filming their athletes' training sessions may use that footage for technique analysis purposes. However, analyzing footage from an entire practice would require manually filtering through video to find instances where swimmers performed each stroke. A machine-learning model that completes this task on video data would allow coaches and swimmers to quickly find training footage of the specific stroke of interest, giving them more time to spend on technique analysis. Additionally, classification models using video data may allow for the automation of disqualifications in competitive racing. Models capable of recognizing the parts of a stroke may be able to flag potentially-illegal movements swimmers make during racing.

To address this problem of stroke classification from video data, we train a machine learning model using data from the `SwimXYZ` data set [9]. This data set contains over 11,000 synthetic videos of swimmers and over 3 million frames for analysis, including variations in camera angle, lighting, motion, and swimmer appearance. We use a subset of this data set to train our model to classify freestyle or breaststroke. We selected 500 instances of videos from a set of freestyle and breaststroke swimmers taken at a water level angle. Each video is considered an instance, and each instance has 16 frames depicting a swimmer. These frames are intended to show the sequence of a swimmer with a frame rate of 15 frames per second. The 500-clip dataset was split into a training set and a validation set, with 80% of the data going to the training set. For training, we randomly sample 5 clips. For testing, we uniformly sample 5 clips. Table 1 summarizes the scope data we used from the `SwimXYZ` data set, and fig. 1 shows a few sample frames from a clip in the full `SwimXYZ` data set.

Label	# clips	# frames	% of dataset
Freestyle	273	4368	54.6
Breaststroke	227	3632	45.4
Total	500	8000	100.0

Table 1: Summary statistics of the subset of the `SwimXYZ` data set. Each clip had frame rate of 15fps and 16 total frames.

Figure 1: The `SwimXYZ` data set contains synthesized videos of figures performing competitive swimming strokes. The above frames from a video depict a swimmer performing backstroke.

4 Solution

Because our problem involves image recognition from pictures, we found it natural to train a convolutional neural network as our classification model. Convolutional neural networks, or CNNs, allow a model to consider spatial data by assuming relationships between attributes. They involve several layers. Convolution layers split attributes into spatial regions and slide special filters over an image to extract information from certain regions. Pooling layers combine information from regions. Fully connected layers act as the standard hidden layers. An output layer makes the final prediction. In this case, the output layer makes the prediction of the stroke being depicted in the image.

Our problem is slightly more complex, as we consider more dimensions than color, height, and width of an image. We also must consider *time*, as our instances are *sequences* of frames that depict several stages of the same stroke. In order to account for this complexity, we utilize a previously established structure of CNN that specializes in classifying types of motion in video data. We trained a CNN using the `MC3_18` model provided by PyTorch [10]. This model is based on the work done by [11] in utilizing CNNs for action recognition. `MC3_18` has 18 layers with a mix of 2D and 3D convolutions. We modify the model slightly by defining a linear layer with 2 output features to match the number of unique labels (breaststroke and freestyle) in our subset of the `SwimXYZ` data set. Besides this final fully connected layer, the model was pretrained, and the convolutional layer parameters are fixed during training. The pre-trained model was built on the kinetics 400 data set, which is a classification task for human activities [10]. We train for 20 epochs and set a batch size of 32. We set the learning rate to be 0.001. The cross-entropy loss function was used for gradient descent training.

Several preprocessing steps of our data were required. Video inputs were treated as 4-dimensional tensors, where the first axis is time, the second axis is color, and the last two axes are spatial (height and width coordinates). For this architecture, instances were normalized to means 0.43216, 0.394666, and 0.37645, and standard deviations 0.22803, 0.22145, and 0.216989, for the colors red, green, and blue, respectively. All instances were rescaled to 128×171 pixels. Training instances were also randomly flipped horizontally with probability 0.5 to help prevent the model from memorizing the training set. Frames were cropped to 112×112 about a random center pixel for compatibility with the predefined model's structure, which involved filter windows based on the assumption of a 112×112 frame size.

5 Outcomes and Evaluation

This section presents the outcomes of our experiments conducted using our slightly modified `MC3_18` CNN model for classifying swimming strokes, specifically freestyle and breaststroke, from video data. We provide an analysis of the model's performance, focusing on key results such as validation accuracy and validation loss trends, and why these results occurred. We discuss the implications of the model's architecture and training setup, and we identify several areas for improvement.

5.1 Performance

The success of our model was evaluated using key performance metrics, including validation accuracy and validation loss. The model achieved a peak validation accuracy at epoch 17, successfully distinguishing between freestyle and breaststroke for 79.5% of the test instances. fig. 2 depicts the model's accuracy over 20 epochs, and fig. 3 depicts the model's loss over those 20 epochs.

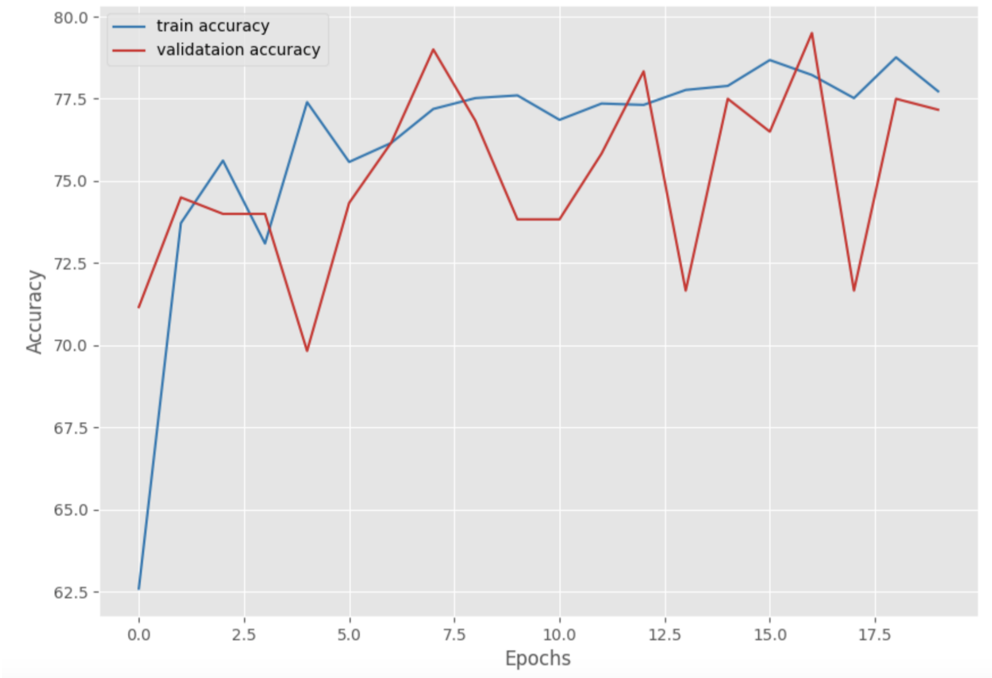

Figure 2: Training and Validation accuracies are depicted for the 20 epochs.

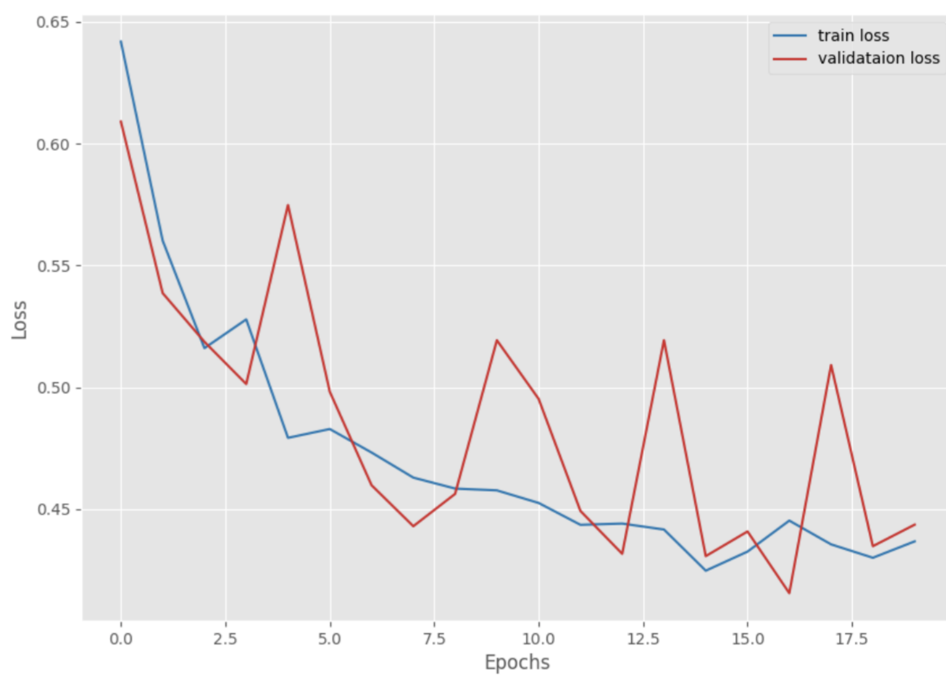

Figure 3: Training and Validation loss are depicted for the 20 epochs.

5.2 Notable Observations

Throughout the training process, the model exhibited steady improvement in both training and validation accuracy, as depicted in fig. 2. This illustrates its capacity to learn and generalize relevant features effectively over time. The consistent decline in validation loss up to epoch 17–as shown in fig. 3–further underscored the model's ability to balance underfitting and overfitting. These findings validate not only the robustness of the model but also the quality of the data set and the overall design of the training pipeline. However, fluctuations in validation metrics beyond epoch 15 suggest areas for further improvement, particularly in addressing overfitting.

5.3 Explanation of Findings

Several factors contributed to the performance of our model. First, the model's 3D convolutional layers played a critical role in capturing the spatial and temporal dynamics inherent in swimming strokes. Unlike 2D CNNs, which are limited to analyzing static frame features, 3D CNNs excel at extracting motion patterns from sequences of frames. This ability is particularly advantageous for recognizing complex and rhythmic movements, such as the coordination of arms and legs in freestyle and breaststroke. The 3D architecture thus made this model an ideal choice for our study.

Second, the balanced data set was instrumental in preventing class bias, ensuring that the model had equal exposure to freestyle and breaststroke examples. Additionally, normalizing the video frames to align with the pre-trained model's expected input format ensured efficient learning. However, one of the key limitations was the use of less than 25% of the available videos in the data set. We also used just 16 frames from each video. Each video in `SwimXYZ` actually lasts 5 seconds with a frame rate of 60fps (300 total frames). Considering this, the 8000 frames we used made up just 0.2% of the data set. This limited data restricted the model's exposure to diverse variations in swimming strokes, which likely contributed to the fluctuations in validation loss and accuracy observed in the later epochs. Despite this limitation, the carefully curated subset enabled the model to generalize effectively within the provided area, highlighting the importance of data quality even in constrained settings.

Lastly, the training strategy itself significantly influenced the model's performance. By using a learning rate of 0.001, the model was able to make gradual yet stable updates to its weights. Freezing the pre-trained layers ensured that the learning process focused on task-specific features, such as stroke dynamics, rather than re-learning basic visual characteristics. The decision to train for 20 epochs struck a balance between learning the stroke dynamics and avoiding overfitting, with the model's performance peaking at epoch 17. However, fluctuations in validation loss after this point indicated that the model began to overfit to the training data, suggesting the potential benefits of additional regularization techniques.

5.4 Meaning of Results

This leads into the implications of this study, which highlights both the strengths and limitations of using 3D CNNs for video-based classification tasks. On one hand, the results reaffirm the potential of this approach for swimming stroke classification, showcasing its ability to extract and analyze complex temporal patterns. On the other hand, the findings point to opportunities for further improvement. Expanding the data set size and diversity could enable the model to capture more subtle stroke variations and reduced overfitting. On top of this, regularization strategies like early stopping could help mitigate overfitting, particularly in the later epochs of training.

In summary, this study demonstrates the effectiveness of the `MC3_18` model for swimming stroke classification while also revealing ways to improve said model. By addressing limitations related to data set size, diversity, and overfitting, future iterations of this approach could achieve even greater accuracy and robustness. The results underscore the potential of 3D CNNs in video-based applications and provide a foundation for further research in motion classification tasks and sports analytics.

6 Conclusions and Future Work

6.1 Summary

Distinguishing between swimming strokes in video is an important problem to solve for the advancement of competitive swimming as a whole. We trained a CNN model to distinguish between two of the four competitive strokes: freestyle and breaststroke. We trained this model using a large dataset of synthetically generated videos depicting swimmers performing breaststroke or freestyle. Our model was structured based on an architecture that showed previous success in classifying human sport movements. The model was able to achieve a modest performance of about 80% validation accuracy at peak performance. Our work faces several limitations, as it was completed with a reduced size data set due to time and computational constraints.

6.2 Suggestions for Future Work

Our project leaves many avenues of future work. Our time and resource constraints limited our model to just two of the four strokes to classify. An area of future work may involve expanding our model to include classification capabilities of the other two competitive strokes: backstroke and butterfly. This is crucial for the deployability of stroke recognition models to real athletes and coaches. Additionally, we only used videos from the side view in our training set. Future work could incorporate the videos taken from above and below the swimmer, which are also included in the `SwimXYZ` data set. In all, our model selected a small portion of the available data for training. Future work could utilize a higher volume of data, which may lead to better prediction accuracy.

Further work on video recognition models specific to swimming environments may increase the performance of stroke classification models. We used the `MC3_18` with few changes; altering this model's underlying structure or developing a separate model specific to swimming motion and aquatic environments could drastically improve performance on this task.

References

[1] T. Koos, "Record-setting world aquatics swimming championships (25m) in budapest caps an incredible year of aquatics," 2024.

[2] S. R. Rath, "Training a video classification model from torchvision," 2023.

[3] J. Blasco, T. M. Chen, J. Tapiador, and P. Peris-Lopez, "A survey of wearable biometric recognition systems," *ACM Comput. Surv.*, vol. 49, Sept. 2016.

[4] G. Brunner, D. Melnyk, B. S. Sigfusson, and R. Wattenhofer, "Swimming style recognition and lap counting using a smartwatch and deep learning," *Proceedings of the 2019 ACM International Symposium on Wearable Computers*, 2019.

[5] E. R. Félix, H. P. d. Silva, B. H. Olstad, J. Cabri, and P. L. Correia, "SwimBIT: A novel approach to stroke analysis during swim training based on attitude and heading reference system (AHRS)," *Sports (Basel)*, vol. 7, Nov. 2019.

[6] H. Fani, A. Mirlohi, H. Hosseini, and R. Herpers, "Swim stroke analytic: Front crawl pulling pose classification," *2018 25th IEEE International Conference on Image Processing (ICIP)*, pp. 4068–4072, 2018.

[7] C. A. Staunton, M. Romann, G. Björklund, and D.-P. Born, "Streamlining performance prediction: data-driven kpis in all swimming strokes," *BMC Research Notes*, vol. 17, Feb. 2024.

[8] J. Xie, J. Xu, C. Nie, and Q. Nie, "Machine learning of swimming data via wisdom of crowd and regression analysis," *Mathematical Biosciences Engineering*, vol. 14, no. 2, pp. 511–527, 2017.

[9] G. Fiche, V. Sevestre, C. Gonzalez-Barral, S. Leglaive, and R. Séguier, "Swimxyz: A large-scale dataset of synthetic swimming motions and videos," in *Proceedings of the 16th ACM SIGGRAPH Conference on Motion, Interaction and Games*, pp. 1–7, 2023.

[10] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Kopf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala, "Pytorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems 32*, pp. 8024–8035, Curran Associates, Inc., 2019.

[11] D. Tran, H. Wang, L. Torresani, J. Ray, Y. LeCun, and M. Paluri, "A closer look at spatiotemporal convolutions for action recognition," 2018.

[12] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and E. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.